



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 03

Objective & Lecture Mapping: In previous labs, we built a Simple Reflex Agent that moved randomly or reacted only to immediate adjacent cells. Today, we transition to a **Goal-Based/Planning Agent**. You will expose the environment's state space to the agent, allowing it to use **Breadth-First Search (BFS)**, **Depth-First Search (DFS)**, and **Uniform-Cost Search (UCS)** to simulate and find paths to the food before taking a physical action.

Part 1: Practical Implementation — Building the Search Agent

Step 1.1: Exposing the World Model

Classical search algorithms require an abstract model of the environment to simulate future states. Currently, your agent is "blind" to the overall map.

1. Create a new Branch called Week_03 in your Forked Github Repo and work on the below Questions.
2. Open `visual_grid_game.py`.
3. Locate the `get_percept(self)` method.
4. Add three new keys to the dictionary to pass the global state to the agent:

```
'grid_size': (self.width, self.height)
'walls': list(self.walls)
'all_food': list(self.food_positions)
```

Step 1.2: Implementing BFS, DFS, and UCS

You will implement three distinct uninformed search strategies. The core node expansion structure is identical; only the data structure for the **Frontier** changes!

1. Open `agent.py` and create a new class called `SearchAgent`.
2. Import required structures: `from collections import deque` and `import heapq`.
3. **BFS:** Create `def bfs_search(...)`. Use a FIFO queue (`deque.popleft()`). This explores the shallowest nodes first.
4. **DFS:** Create `def dfs_search(...)`. Use a LIFO stack (`list.pop()`). This explores the deepest nodes first.
5. **UCS:** Create `def ucs_search(...)`. Use a Priority Queue (`heapq.heappop()`) ordered by the total path cost $g(n)$.
6. For all three algorithms, you must maintain a `reached` set (or visited array) to track explored states. This converts your algorithm from a *Tree Search* to a *Graph Search*, preventing infinite loops.

Step 1.3: Executing and Comparing Offline Plans

The agent must now form a complete plan and execute it step-by-step.

1. In your `SearchAgent.__init__`, add an empty list: `self.plan = []` and a config string `self.active_algo = 'BFS'`.
2. In `sense_and_act(self, percept)`, check if `self.plan` is empty.
3. If empty, find the closest food pellet from `percept['all_food']`, execute the search method matching `self.active_algo`, and store the resulting sequence of actions in `self.plan`.
4. Return the first action from the plan using `return self.plan.pop(0)`.
5. **Observation Task:** Change `self.active_algo` between 'BFS', 'DFS', and 'UCS' and run the simulation. Watch how DFS takes erratic, winding paths, while BFS and UCS take direct, optimal paths!

Part 2: Theoretical Evaluation

Based on your implementation and Lecture 04, complete the following analytical questions.

1. (Remember) You built your search logic based on the environment coordinates. What is the fundamental difference between the "State Space" and the "Search Tree"?

The **state space** is the complete set of all possible states or positions in the environment. The **search tree** is the structure created by the search algorithm while exploring those states. The same state may appear more than once in a search tree if repeated states are not controlled.

2. (Understand) In Step 1.2, you were instructed to use a `reached` set. In graph search theory, what specific problem does this set solve, and what would happen to your DFS agent without it?

The reached set stores the states that have already been visited. It prevents the agent from exploring the same state again and again. Without it, DFS may enter a loop, repeatedly visit the same positions, and may never reach the goal.

3. (Analyze) Compare the visual paths generated by BFS and DFS in your simulation. Why is BFS guaranteed to be optimal in this specific grid, whereas DFS produces winding, suboptimal routes?

BFS explores the shallowest nodes first. In our grid, every move has the same cost, so the path with the fewest moves is also the lowest-cost path. Therefore, BFS finds an optimal path. DFS explores one path deeply before checking other paths, so it may take a long and winding route even when a shorter path exists.

4. (Evaluate) If we scaled `visual\_grid\_game.py` up to a massive 1000x1000 grid, BFS and UCS might crash before finding food. According to the time and space complexity formulas from the lecture, contrast the memory bottlenecks of BFS versus DFS.

BFS stores many nodes from each search level in memory, so its space complexity is approximately $O(b^d)$. This can require a very large amount of memory in a huge grid. DFS mainly stores the current path and some unexplored nodes, so its space complexity is approximately $O(bm)$. Therefore, DFS usually uses much less memory than BFS.

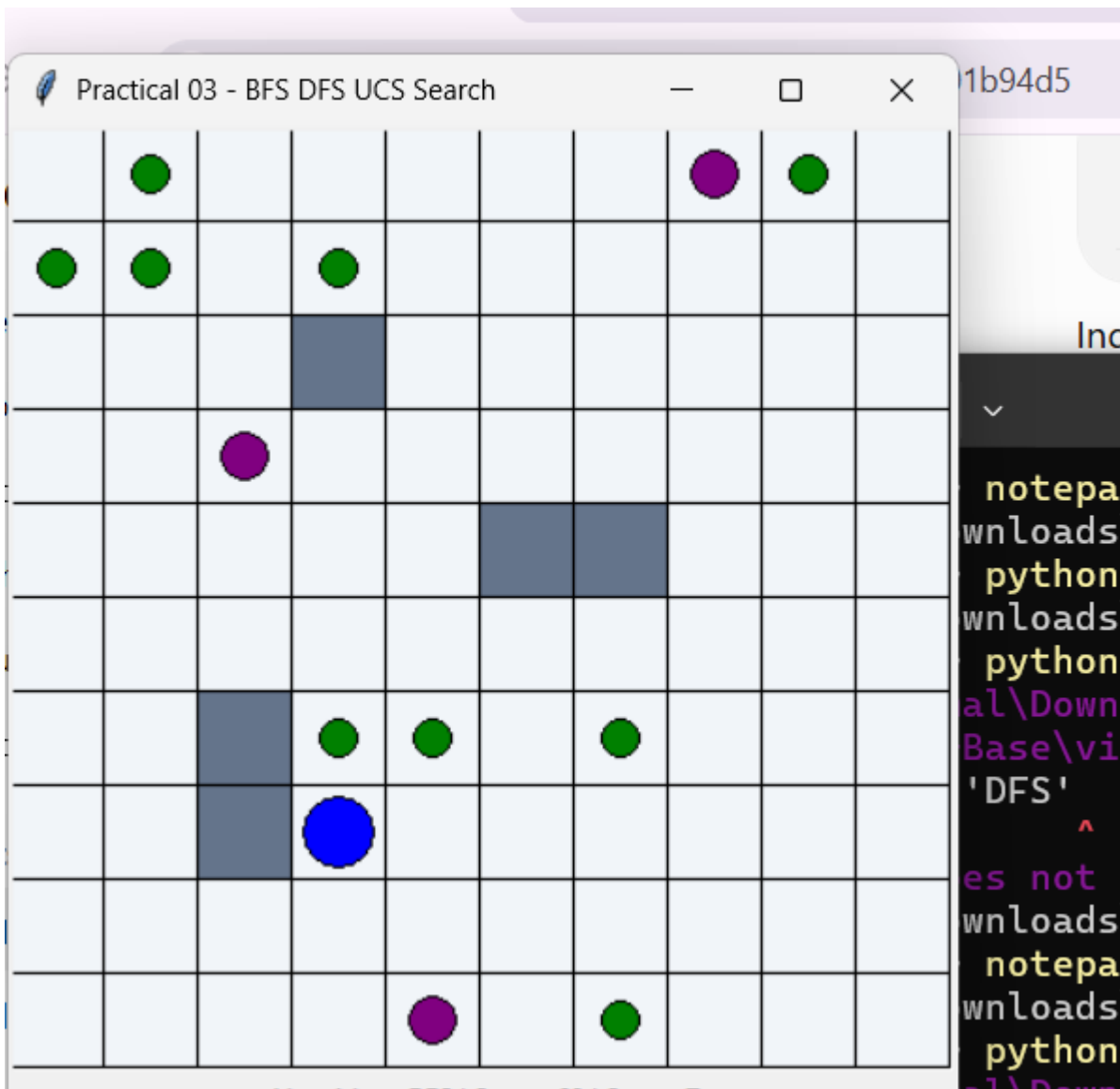
Once Completed Get a PDF of the completed File and Push into the Week 03 brach in the Github Project

Finalize and Lock Lab 03 Documentation

Incomplete: 0/11 questions answered. Please provide detailed responses for all questions.

**FACULTY OF COMPUTING**

BFS



DFS

Practical 03 - BFS DFS UCS Search

Finished! Algorithm: DFS | Score: 65 | Steps: 60

Python

app = GridGameGUI(root)

```
ts, 'Right', 'Right', 'Right', 'Right', 'Right', 'Right', 'Up', 'Left', 'Left',
'Left', 'Left', 'Up', 'Right', 'Right', 'Right', 'Right', 'Right', 'Right', 'Rig
ft', 'Left', 'Left', 'Left', 'Left', 'Left', 'Left', 'Up', 'Right', 'Rig

t', 'Up', 'Up', 'Up', 'Up']
ds\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
bad visual_grid_game.py
ds\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
on visual_grid_game.py
ds\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
on visual_grid_game.py
nloads\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---
visual_grid_game.py", line 148
t match any outer indentation level
ds\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
bad visual_grid_game.py
ds\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
Base\IT3012---Practical-Base> python visual_grid_game.py
File "C:\Users\Mohamed Ashmal\Downloads\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---
ical-Base\IT3012---Practical-Base\visual_grid_game.py", line 145
self.agent = SearchAgent()
IndentationError: unindent does not match any outer indentation level
PS C:\Users\Mohamed Ashmal\Downloads\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
Base\IT3012---Practical-Base> python -m py_compile visual_grid_game.py
PS C:\Users\Mohamed Ashmal\Downloads\IT3012---Practical-Base-week_04\IT3012---Practical-Base-week_04\IT3012---Prac
Base\IT3012---Practical-Base> python visual_grid_game.py
```

Practical 03 - BFS DFS UCS Search

Algorithm: UCS | Score: 140 | Steps: 15

PowerShell

notepad visu

```
' , 'Right', 'Up', '
\Downloads\IT3012--
se> notepad visual_
\Downloads\IT3012--
se> python visual_g
\Downloads\IT3012--
se> python visual_g
shmal\Downloads\IT3
al-Base\visual_grid
= 'DFS'
does not match any
\Downloads\IT3012--
se> notepad visual_
Base\IT3012---Practical-Base> python visual_g
```